COMPONENTS:	ORIGINAL MEASUREMENTS:
(1) Scandium nitrate; $Sc(NO_3)_3$; [13465-60-6] (2) 2-Octanol (sec caprylic alcohol); $C_8H_{18}O$; [123-96-6] (3) Water ; H_2O ; [7732-18-5]	Pushkina, G. Ya.; Komissarova, L.N. *Zh. Neorg. Khim.* 1963, *8*, 1498-504; *Russ. J. Inorg. Chem. Engl. Transl.* 1963, *8*, 777-81.
VARIABLES: One temperature: 25°C	PREPARED BY: T. Mioduski and M. Salomon

EXPERIMENTAL VALUES:

Composition of saturated solution at 25°C

Sc_2O_3	Sc	NO_3	H_2O[a]	$CH_3(CH_2)_5CH(OH)CH_3$[a]	$Sc(NO_3)_3$[a]	
mass %	mass %	mass %	mass %	mass %	mass %	mol kg^{-1}
9.06	5.91	24.30	9.47	60.18	30.35	1.886

a. Calculated by the compilers based on Sc_2O_3 mass %, and using 1977 IUPAC recommended atomic weights.

AUXILIARY INFORMATION

METHOD/APPARATUS/PROCEDURE:

Isothermal method used. The initial materials were placed in glass vessels and thermostated. Equilibrium was reached within 48 days as ascertained by successive analyses. 0.1 - 0.5 gram samples were removed for analyses using a No. 2 or No. 3 Schott filter fitted to a pipet. 0.5-1 g of solid was also removed for analyses.

Scandium was determined gravimetrically as the oxide. The organic material was first destroyed by heating HNO_3 to dryness; the residue was then dissolved in HCl, $Sc(OH)_3$ precipitated and ignited to the oxide. Nitrogen was determined by Devarda's method, and water was calculated from the loss in weight upon heating to 900°C. The results given above are the mean of 3-4 determinations. The Sc_2O_3:N_2O_5 molar ratio in the liquid and solid phases is constant at approximately 1:3.

SOURCE AND PURITY OF MATERIALS:

$Sc(NO_3)_3 \cdot 4H_2O$ was crystallized from HNO_3 sln containing at least 44.26 mass % N_2O_5. Chemical analyses resulted in the following (mean of 3 detns): Sc 14.84 mass %, NO_3 61.33 mass %. Impurities in the salt stated not to exceed 0.01 %. Prior to use the salt was dried in a desiccator over H_2SO_4 to remove hydroscopic water.

Source of the organic solvent not specified, but authors state that its purity was checked optically.

ESTIMATED ERROR:

Soly: reproducibility around ± 1% (compilers).

Temp: accuracy ± 0.1 K (authors).

REFERENCES: